

Group 4

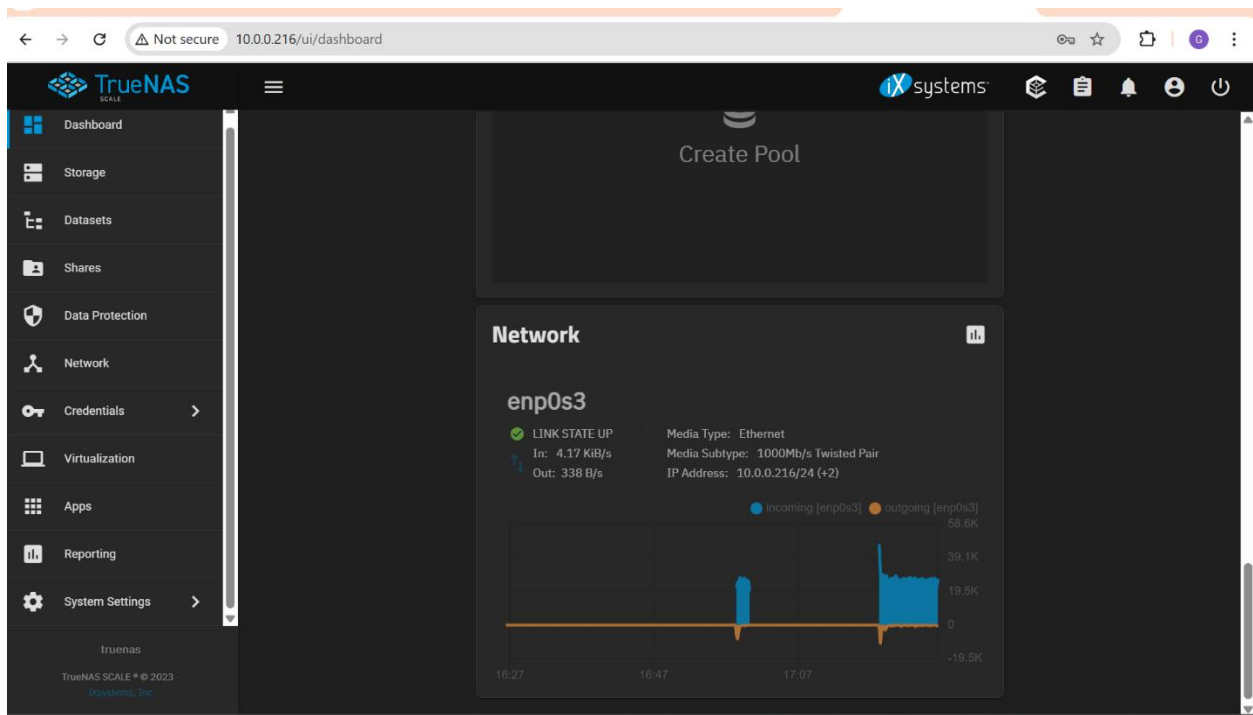
Local IP address showing logged in

```
TrueNAS_25 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[13206.014501] clocksource: Switched to clocksource kvm-clock
[13207.794628] systemd[1]: sysstat-collect.service: Deactivated successfully.
[13207.795938] systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
[13208.449191] systemd-coredump[6932]: Process 6169 (systemd-journal) of user 0 dumped core.
[13208.449897] systemd-coredump[6932]: Core dump diverted to /var/lib/systemd/coredump/core.systemd-journal.0.dbdc7ce32750402786d9a478d65ae8ff.6169.1767761130000000.zst
[13208.451137] systemd-coredump[6932]: Stack trace of thread 6169:
[13208.451838] systemd-coredump[6932]: #0 0x00007f78a771b04e __libc_read (libpthread.so.0 + 0x1204e)
[13208.453166] systemd-coredump[6932]: #1 0x0000560ed6e4a8f3 n/a (systemd-journald + 0x138f3)
[13208.453935] systemd-coredump[6932]: #2 0x00007f78a79a35f0 n/a (libsystemd-shared-252.so + 0x23c5f0)
[13208.455346] systemd-coredump[6932]: #3 0x00007f78a79a38fd sd_event_dispatch (libsystemd-shared-252.so + 0x23c8fd)
[13208.456864] systemd-coredump[6932]: #4 0x00007f78a79a64d0 sd_event_run (libsystemd-shared-252.so + 0x23f4d0)
[13208.458534] systemd-coredump[6932]: #5 0x0000560ed6e3f54d n/a (systemd-journald + 0x854d)
[13208.469702] systemd-coredump[6932]: #6 0x00007f78a7557d0a __libc_start_main (libc.so.6 + 0x23d0a)

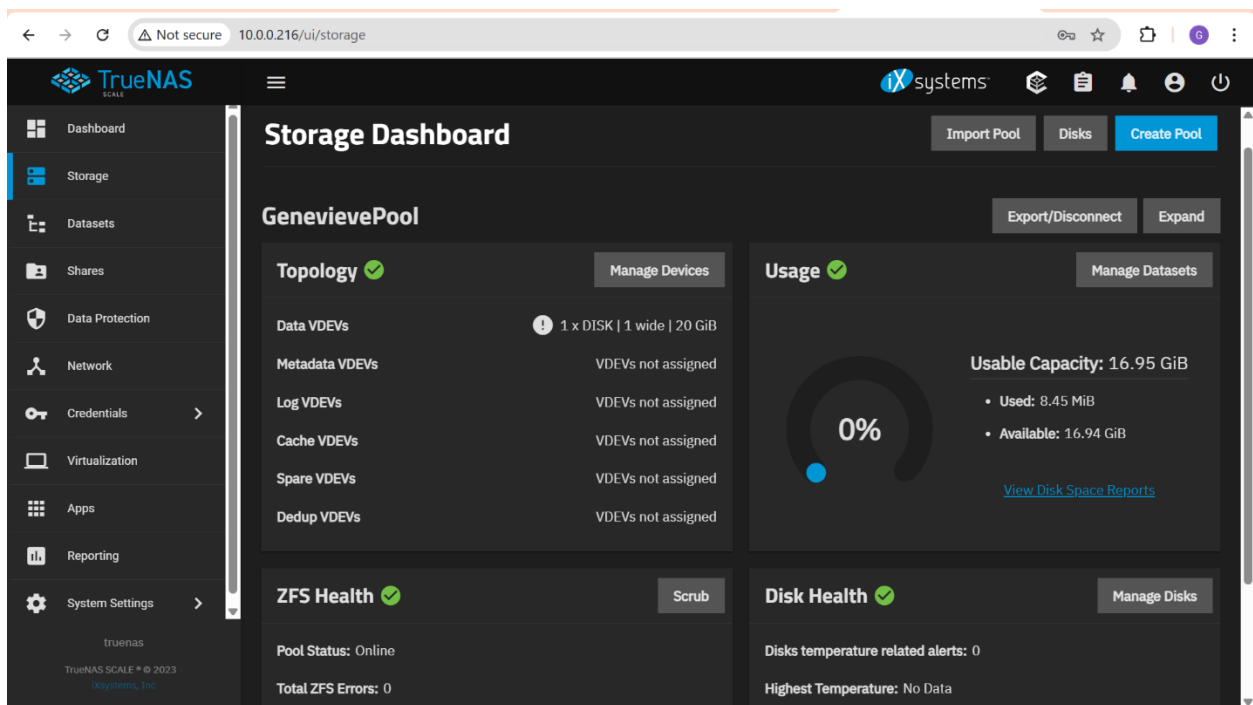
The web user interface is at:
http://10.0.0.216
http://[2604:3d09:d088:7100:a00:27ff:feed:c918]
https://10.0.0.216
https://[2604:3d09:d088:7100:a00:27ff:feed:c918]

1) Configure network interfaces
2) Configure network settings
3) Configure static routes
4) Change local administrator password
5) Reset configuration to defaults
6) Open TrueNAS CLI Shell
7) Open Linux Shell
8) Reboot
9) Shutdown

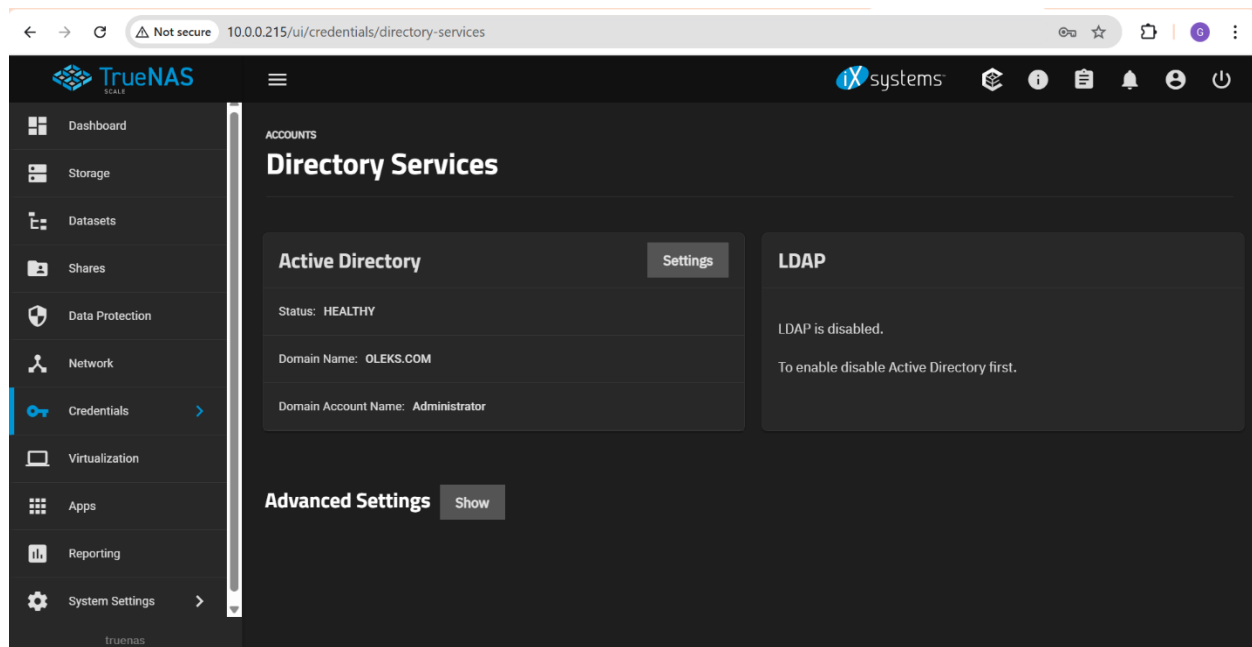
Enter an option from 1-9: _
```



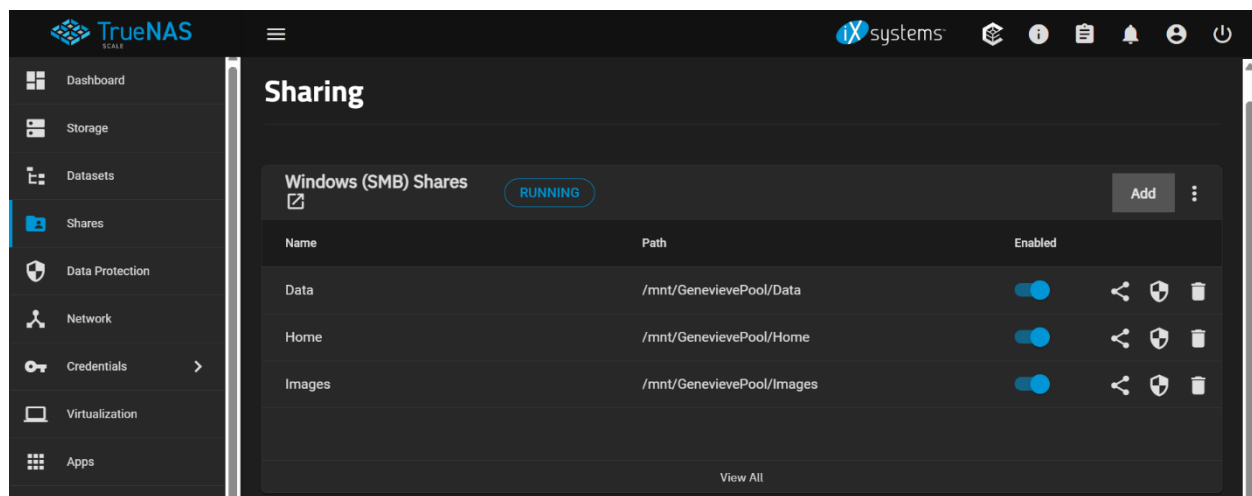
Second Disk added to TrueNAS installation



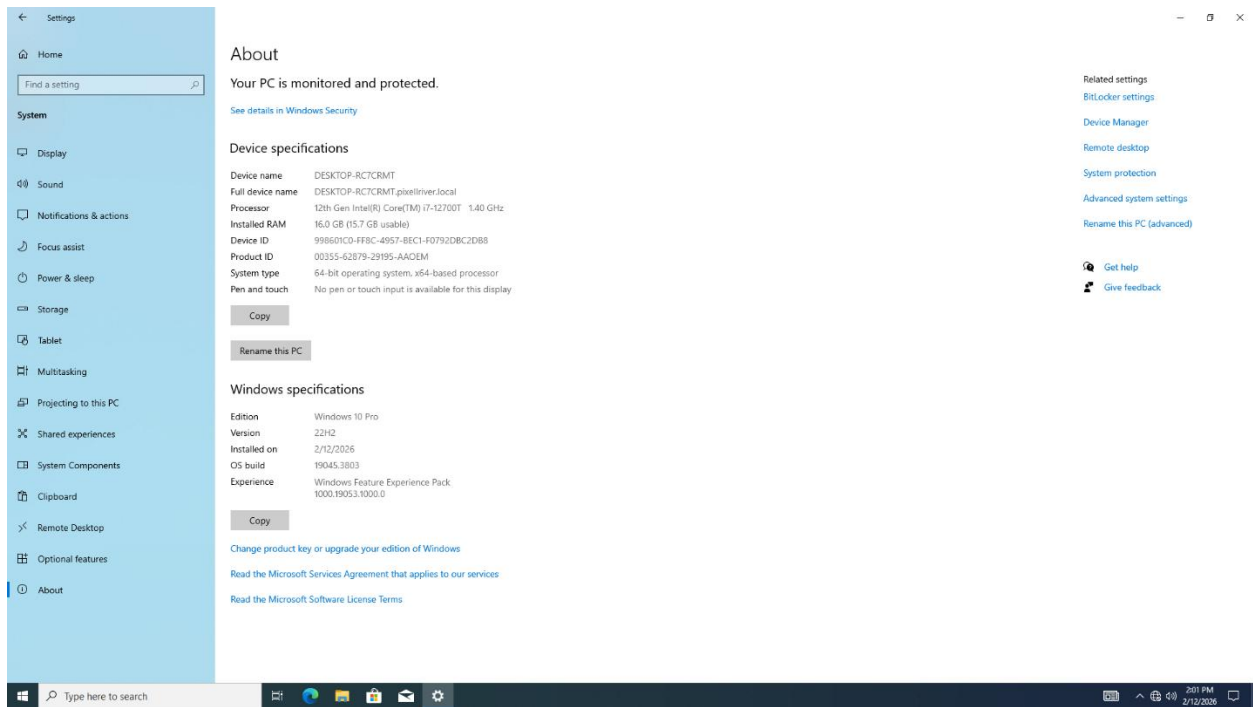
Joined TrueNAS Server to Active Directory Successfully



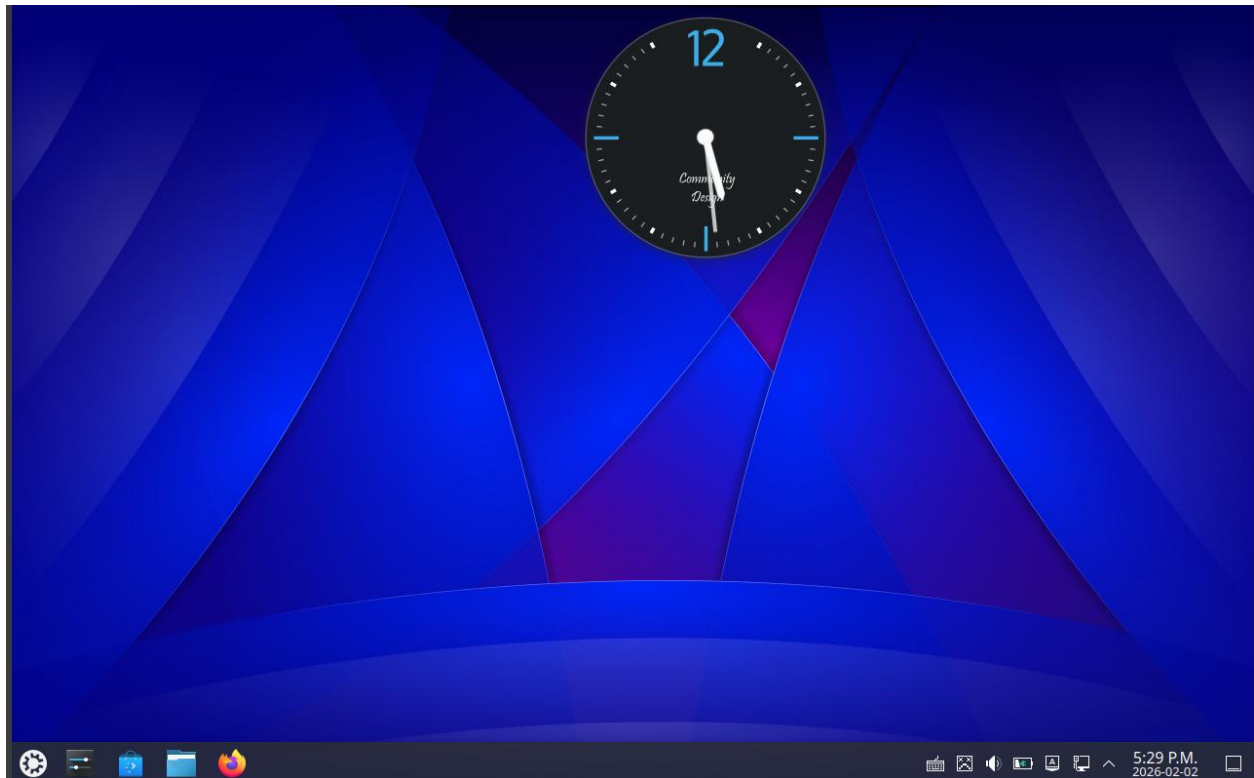
Network Shares and Permissions Configuration



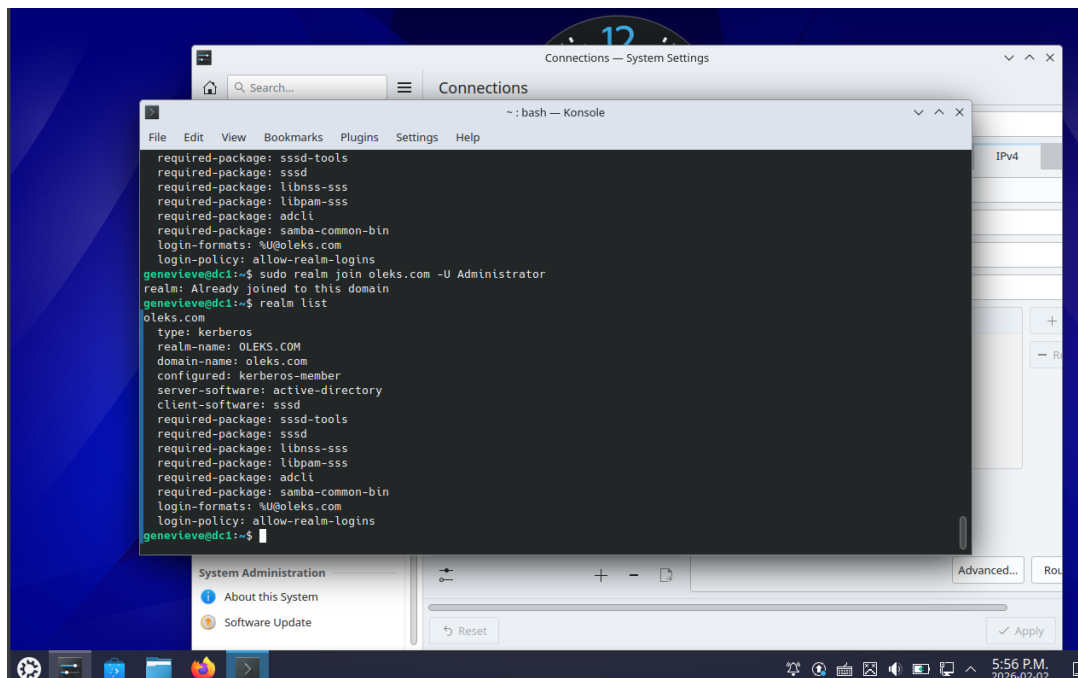
Successfully configured and Installed Windows Client and joined AD Domain



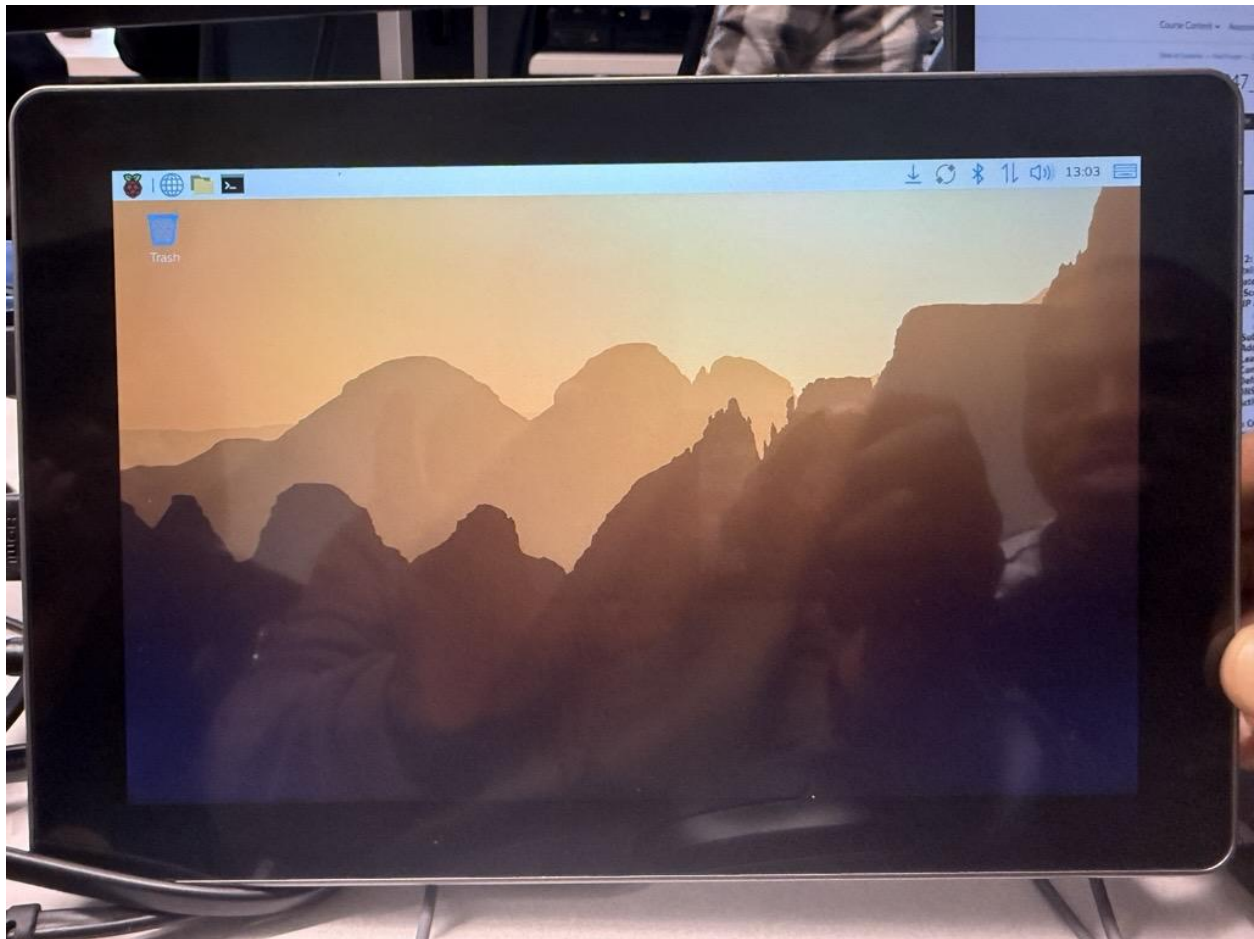
Linux client Installation and Configuration



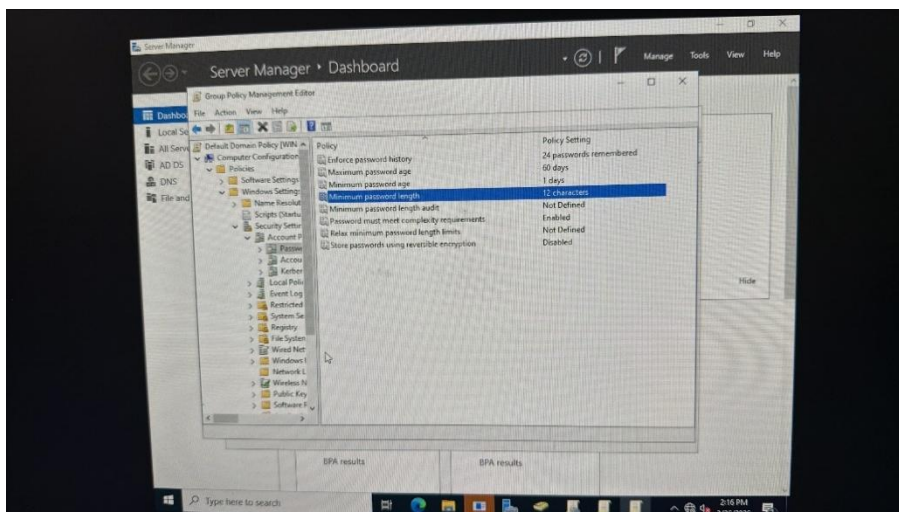
Successfully Joined Linux to AD Domain

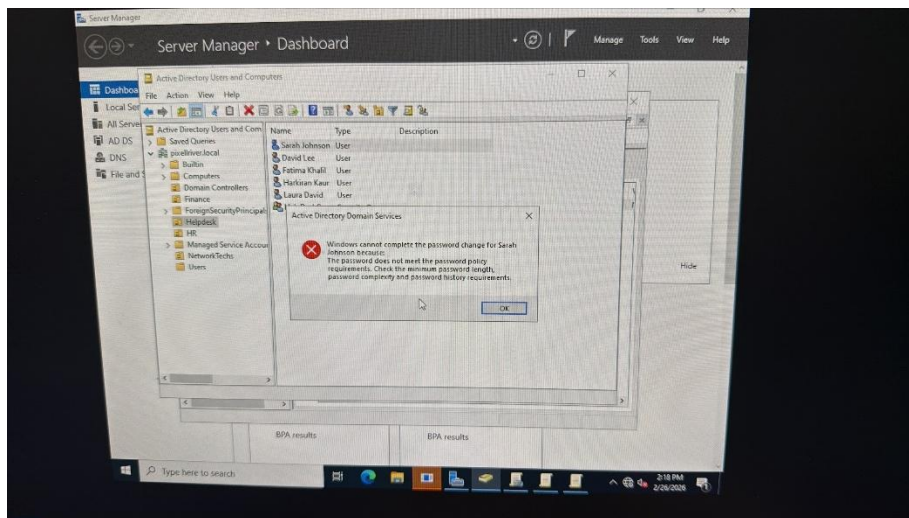


Successfully completed the Raspberry Pi setup process.

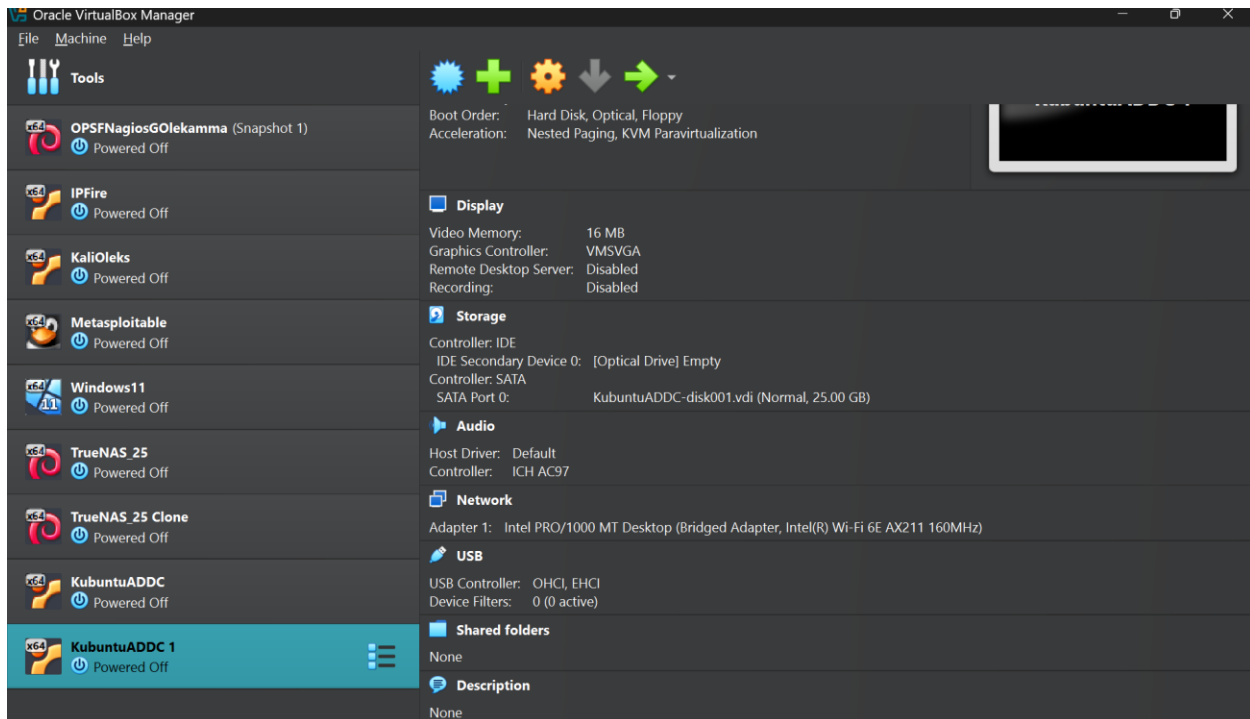


Successfully created and configured new Group Policies without errors.





Completed an appliance creation in Virtual Box



Ansible Inventory Files and Playbooks and Software Deployment with Ansible

```
admin@kubuntu01:~$ ssh-keygen -t rsa -b 4096
Bad escape character 'ygen'.
admin@kubuntu01:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
```

```
Enter file in which to save the key (/home/admin/.ssh/id_rsa): Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/admin/.ssh/id_rsa
Your public key has been saved in /home/admin/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:NrHvHZhSJ6LFFJyhPRJWncGv8EM91vhbUHok8LSWncI admin@kubuntu01
The key's randomart image is:
```

```
+---[RSA 4096]-----+
|
|  oo+=.0... |
| . +0.+ .0.++|
| o = o E=*.* |
| +.+ *.*= . |
| S+oo.o o |
| + =+= . . |
| . . +.. o |
| o . . . |
| . . . |
+-----[SHA256]-----+
```

```
admin@kubuntu01:~$
```

~ : bash — Konsole

File Edit View Bookmarks Plugins Settings Help

```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.122.247' (ED25519) to the list of known hosts.
admin@192.168.122.247's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-27-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

Expanded Security Maintenance for Applications is not enabled.

```
324 updates can be applied immediately.
196 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

```
38 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
admin@kubuntu03:~$ exit
```

```

admin@kubuntu01:~$ ssh admin@192.168.122.5
The authenticity of host '192.168.122.5 (192.168.122.5)' can't be established.
ED25519 key fingerprint is SHA256:dPwVR7kFiLSPRQJAgwW8KdLxIzk4sC00WBFQrnH5q68.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.122.5' (ED25519) to the list of known hosts.
admin@192.168.122.5's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

76 updates can be applied immediately.
41 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

38 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Mon Mar 23 16:53:04 2026 from 192.168.122.5
admin@kubuntu02:~$ exit
logout
Connection to 192.168.122.5 closed.

```

```

admin@kubuntu01:~/test-project$ ansible-playbook -i hosts.ini playbook1.yml

PLAY [Install net-tools on app servers] *****

TASK [Gathering Facts] *****
[WARNING]: Host 192.168.122.5 is using the discovered Python Interpreter at '/usr/bin/python3.12', but future installation of another Python Interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.122.5]

TASK [Ensure net-tools is installed] *****
changed: [192.168.122.5]

PLAY [Check network configuration with ifconfig] *****

TASK [Gathering Facts] *****
ok: [192.168.122.5]

TASK [Run ifconfig command] *****
changed: [192.168.122.5]

TASK [Display ifconfig output] *****
ok: [192.168.122.5] => {
  "ifconfig_output.stdout_lines": [
    "ens160: flags=4096<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500",
    "    inet 192.168.122.5 netmask 255.255.255.0 broadcast 192.168.122.255",
    "    inet6 fe80::7d9c:2471:109c:217b prefixlen 64 scopeid 0x20<link>",
    "    ether 52:54:00:12:34:56 txqueuelen 1000 (Ethernet)",
    "    RX packets 481693 bytes 544920589 (544.9 MB)",
    "    RX errors 0 dropped 30 overruns 0 frame 0",
    "    TX packets 100025 bytes 8981586 (8.9 MB)",
    "    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0",
    "",
    "lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536",
    "    inet 127.0.0.1 netmask 255.0.0.0",
    "    inet6 ::1 prefixlen 128 scopeid 0x10<host>",
    "    loop txqueuelen 1000 (Local Loopback)",
    "    RX packets 1059 bytes 184664 (184.6 KB)",
    "    RX errors 0 dropped 0 overruns 0 frame 0",
    "    TX packets 1059 bytes 184664 (184.6 KB)",
    "    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0"
  ]
}

PLAY RECAP *****
192.168.122.5      : ok=5  changed=2  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

```
admin@kubuntu01:~/test-project$ ansible-playbook -i hosts.ini install_passlib.yml

PLAY [Install passlib on app servers] *****

TASK [Gathering Facts] *****
[WARNING]: Host '192.168.122.5' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.122.5]

TASK [Install passlib via pip] *****
ok: [192.168.122.5]

PLAY RECAP *****
192.168.122.5      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

admin@kubuntu01:~/test-project$
```

```
admin@kubuntu01:~/test-project$ ansible-playbook -i hosts.ini create_user.yml

PLAY [Create user sara on app servers] *****

TASK [Gathering Facts] *****
[WARNING]: Host '192.168.122.5' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.122.5]

TASK [Create HR group] *****
changed: [192.168.122.5]

TASK [Create user sara] *****
changed: [192.168.122.5]

PLAY RECAP *****
192.168.122.5      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

admin@kubuntu01:~/test-project$
```

```
admin@kubuntu01:~/test-project$ ansible multi -m file -a "src=/tmp/test/ansible.cfg dest=/tmp/test/ansible_symlink.cfg state=link"
[WARNING]: Host '192.168.122.5' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.122.5 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "dest": "/tmp/test/ansible_symlink.cfg",
  "gid": 1001,
  "group": "ansible",
  "mode": "0777",
  "owner": "ansible",
  "size": 21,
  "src": "/tmp/test/ansible.cfg",
  "state": "link",
  "uid": 1001
}
[WARNING]: Host '192.168.122.247' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.122.247 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "dest": "/tmp/test/ansible_symlink.cfg",
  "gid": 1001,
  "group": "ansible",
  "mode": "0777",
  "owner": "ansible",
  "size": 21,
  "src": "/tmp/test/ansible.cfg",
  "state": "link",
  "uid": 1001
}
}
```

